

Developers Mania

Working Group

Alfayo Nelson Central Command

Software Requirements Specification

Data-driven campaign management and voter mobilisation platform for the Nyali Constituency parliamentary candidacy. Three integrated applications — Central Command (web), Field App (mobile), and NyaliTrack (election day) — sharing one secure backend.

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0. Problem Statement & Proposed Solution

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This section justifies everything that follows. It establishes why the system exists before specifying what it does.

0.1 Problem Statement

0.1.1 Problem Context

Kenyan parliamentary campaigns at the constituency level operate on incomplete, fragmented, and largely informal intelligence. A typical campaign in a competitive coastal constituency such as Nyali manages relationships with hundreds of community leaders across five wards, plans hundreds of activities over a fourteen-month cycle, and on election day deploys two to four polling agents across approximately one hundred polling stations. Most of this work is coordinated through WhatsApp groups, paper notebooks, ad-hoc spreadsheets, and the memories of individual coordinators. Critical institutional knowledge — which imam is supportive, which boda stage chairman has flipped, what issue a particular village cares about — exists only in conversations and rarely survives team turnover.

Nyali Constituency comprises five wards (Kadzandani, Kongowea, Mkomani, Frere Town, and Ziwa La Ng'ombe), approximately one hundred polling stations, and roughly 130,000 registered voters distributed across formal sub-locations and dozens of informally bounded villages and estates. The candidate, Alfayo Nelson, requires a system that aggregates ground-level intelligence into a coherent strategic picture, supports daily field operations across the five wards, and on 9 August 2027 enables real-time polling station operations and turnout coordination.

0.1.2 Problem Statement (Core)

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The four-part structure makes the problem statement testable.

The problem of

fragmented, undocumented, and individually held campaign intelligence and field operations data

affects

parliamentary campaigns operating in competitive constituencies such as Nyali — specifically the candidate, campaign manager, strategists, ward coordinators, polling station leads, polling agents, canvassers, and influence liaisons who together must coordinate voter mobilisation across five wards and one hundred polling stations.

the impact of which is

loss of institutional knowledge when team members leave; duplicated and contradictory field work; mis-targeted candidate time and resources; poor coverage in critical villages; reactive rather than data-driven strategy; preventable failures on election day including missing agent deployments, slow turnout response, and unverified incident reports; and post-election inability to maintain constituency relationships for a future term.

A successful solution would

give the candidate and campaign leadership a single source of truth for every village, community leader, community site, issue, activity, and team member in the constituency; provide field coordinators with a fast, offline-capable mobile tool to capture ground-truth data as it happens; enable real-time polling station operations on election day with paper and SMS fallbacks; and produce an institutional asset that supports the candidate's relationship with Nyali through a full parliamentary term and into the 2032 cycle.

0.1.3 Evidence of the Problem

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Observed pain points drawn from product owner interviews and review of comparable Kenyan parliamentary campaigns.

- Critical community-leader contact lists exist primarily in coordinators' phones and personal WhatsApp chats; coordinator turnover routinely results in losing access to entire ward networks.
- No central record exists of where the candidate or surrogates have visited; visits are duplicated to high-profile sites while several villages receive no contact for months.
- Village-level issues raised in barazas are noted by individual coordinators but rarely captured in a form the candidate can reference before a return visit.
- Polling-station-level results from 2013, 2017, and 2022 are publicly available but have never been integrated with the campaign's field assessments, preventing targeted persuasion strategy.
- Election-day operations in past campaigns have depended on dozens of parallel WhatsApp groups; aggregating turnout data and incident reports across one hundred stations has been manual and slow.
- Voter register data and demographic intelligence are typically held in spreadsheets without access controls, creating compliance exposure under the Kenya Data Protection Act, 2019.

0.1.4 Root Cause Analysis

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Symptoms are what stakeholders complain about. Root causes are why those symptoms exist.

Symptom (Observable Problem)	Root Cause
Community leader relationships lost when coordinators leave	No structured shared system for leader contacts and relationship status; data lives in individuals' phones
Candidate visits repeat in some areas while villages go uncovered	No central record of visits or coverage; planning is based on memory and personal networks
Village issues raised at barazas are forgotten before follow-up	No structured issue capture at the point of conversation; reliance on paper notes or memory
Strategy is reactive rather than targeted	No integrated view of historical results, demographics, coverage, and influence; analysis happens in separate spreadsheets
Election-day operations are slow and error-prone	Coordination through ungoverned WhatsApp groups; no structured reporting, no real-time roll-up, no audit trail
Voter and contact data not handled compliantly	No access controls; no audit logging; no Office of the Data Protection Commissioner registration

0.2 Proposed Solution

0.2.1 Solution Overview

Alfayo Nelson Central Command is a three-application campaign platform sharing one PostgreSQL/PostGIS database. The Central Command web application gives the candidate, campaign manager, strategists, and constituency coordinator a single integrated view of geographic, demographic, electoral, community, and operational intelligence across Nyali. The Field App, a native mobile application built with Expo and React Native, gives ward coordinators, polling station leads, agents, canvassers, and influence liaisons a fast, offline-capable tool to capture ground-truth data and receive assignments. NyaliTrack, a dedicated election-day application sharing the same backend, supports real-time polling station operations, agent check-ins, turnout tracking, and incident management on 9 August 2027.

0.2.2 How It Solves the Problem

Problem Element	How Alfayo Nelson Central Command Addresses It
Loss of community-leader networks on turnover	Structured leader directory with ownership, relationship temperature, last-contact tracking, and ward-coordinator review queue for new entries
Duplicated visits and uncovered villages	Visit logging in three taps in the Field App; three heatmaps in Central Command (Coverage, Influence Density, Persuasion Opportunity) showing where to deploy next
Forgotten village-level issues	Issue capture at point of conversation with controlled vocabulary; ward and village rollups surface the issues the candidate must address
Reactive, ungrounded strategy	Integration of 2013/2017/2022 polling-station results, demographics, field assessments, and live coverage data into a single Pollings & Analysis tab
Slow, error-prone election-day operations	NyaliTrack: real-time station check-ins, structured incident reports, paper and SMS fallbacks for network failures, full audit trail
Compliance exposure on voter data	Office of the Data Protection Commissioner registration, row-level security, column-level encryption of sensitive fields, comprehensive audit logging

0.2.3 Value Proposition

For Kenyan parliamentary campaigns operating in competitive constituencies who struggle with fragmented and individually held campaign intelligence, Alfayo Nelson Central Command is a three-application campaign platform that consolidates geographic, community, electoral, and operational data into one secure system. Unlike spreadsheet-and-WhatsApp coordination, the platform produces durable institutional memory, supports offline field work, complies with the Data Protection Act, and outlasts a single election cycle.

0.2.4 MVP Scope Rationale

MVP Includes:

- Geographic and demographic intelligence at constituency, ward, sub-location, village, and polling-station levels
- Community leader and community site directories with field-app data entry and review queues
- Village-level issue tracking with controlled vocabulary
- Campaign activity and event planning with attendance and outcome tracking
- Visit logging with offline support and three heatmap visualisations
- Meeting coordination with automated WhatsApp and email notifications
- Pollings & Analysis dashboards covering electoral history, demographics, race intelligence, and campaign performance
- Role-based access with row-level enforcement and comprehensive audit logging
- NyaliTrack election-day operations module — real-time polling station status, turnout pace, incident management
- Bilingual English and Swahili interface

MVP Excludes:

- Social media monitoring or sentiment analysis — outside scope of voter mobilisation
- AI-generated campaign content — significant governance and data exposure concerns
- Financial accounting and campaign expenditure tracking — handled by a separate system for audit reasons
- Direct integration with IEBC live results infrastructure — IEBC does not publish such an API; results in NyaliTrack are agent-reported
- Voter registration assistance workflows — IEBC mandate, not the campaign's
- Multi-constituency deployment — the platform is designed to be reusable but only Nyali is in this MVP

Core Hypothesis Being Tested:

A constituency-level campaign that operates from a shared institutional-memory system through the full fourteen-month campaign cycle will demonstrate measurably better village coverage, faster election-day operations, and more durable post-election constituency relationships than a campaign coordinated through ad-hoc spreadsheets and WhatsApp groups.

1. Introduction

1.1 Purpose

This Software Requirements Specification defines the complete functional and non-functional requirements for Alfayo Nelson Central Command v1.0. It serves as the authoritative product document governing design, development, testing, and validation across the three integrated applications (Central Command web, Field App mobile, NyaliTrack election-day app). The intended audience is the Developers Mania project team, the Alfayo Nelson Campaign leadership, the campaign's engaged legal counsel, and any third party conducting security audit or compliance review of the platform.

1.2 Scope

SCOPE RULE

The Out-of-Scope list below is mandatory and non-negotiable. It is the platform's primary defence against scope creep over the fourteen-month build cycle.

In Scope — v1.0

- Three integrated applications sharing one backend: Central Command (Next.js web), Field App (Expo mobile), NyaliTrack (Expo mobile, election day)
- Geographic intelligence: constituency, ward, sub-location, village, and polling-station data and visualisations for Nyali
- Community intelligence: leaders, sites (mosques, churches, schools, markets, boda stages, chamas, halls), and influence networks
- Village-level issue tracking with controlled vocabulary
- Campaign activity planning, execution, and outcome capture
- Visit logging with offline-first support; three heatmap visualisations
- Committed Supporter Network — tracked list of voters with documented support commitments arising from community programs, with strict access controls and audit logging
- Meeting coordination with automated WhatsApp and email notifications to configured stakeholders
- Pollings & Analysis dashboards covering 2013, 2017, 2022 elections and current race intelligence
- Role-based access control enforced at the database level via PostgreSQL row-level security
- Comprehensive audit logging meeting Kenya Data Protection Act, 2019 requirements
- Office of the Data Protection Commissioner registration support documentation
- Bilingual English/Swahili interfaces
- Election-day operations: real-time polling station check-ins, turnout reports, incident management, paper and SMS fallbacks
- Countdown widget to election date, configurable in administrator settings

Out of Scope — v1.0

- Social media monitoring, scraping, or sentiment analysis
- AI-generated campaign content, speeches, or messaging
- Financial accounting, expenditure tracking, donor management, or M-Pesa payment processing
- Direct integration with the IEBC live results infrastructure (IEBC publishes no such API)

- Voter registration assistance, polling card distribution, or eligibility checking workflows (these are IEBC mandates)
- Multi-constituency or multi-tenant deployment in this release; the database supports future extension
- Public-facing voter outreach channels (e.g. candidate website, mass-SMS to voters at registration time)
- Opposition research beyond publicly available information; strategic notes are restricted to four roles and never aggregated
- Facial recognition, biometric voter identification, or any technology not permitted under Kenyan electoral law
- Integration with the National Police Service or other government databases

1.3 Definitions and Abbreviations

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Every domain term, role, and prefix used in this document is defined here. Reviewers must verify no acronym appears in the document without a definition.

Domain Terms

Term	Definition
Aspirant	The candidate seeking election. In this document, Alfayo Nelson, parliamentary candidate for Nyali Constituency.
Baraza	Public community meeting, typically called by an elder or political figure for open discussion of local issues.
Boda boda	Motorcycle taxi. Boda stages are organised groupings of riders, each typically with a chairman who serves as a political influencer.
Chama	Informal Swahili term for a savings group, investment club, or self-help group; politically significant in Kenyan communities.
Constituency	Electoral unit electing a Member of Parliament. Nyali is one of six constituencies in Mombasa County.
D-Day	Election Day. For this release, 9 August 2027.
IEBC	Independent Electoral and Boundaries Commission, Kenya's electoral management body.
KNBS	Kenya National Bureau of Statistics.
MCA	Member of County Assembly; ward-level elected representative.
MP	Member of Parliament; constituency-level elected representative.
Mtaa	Informal Swahili term for a village, estate, or neighbourhood within a ward; the geographic unit residents identify with.
ODPC	Office of the Data Protection Commissioner, the regulator for the

Term	Definition
	Kenya Data Protection Act, 2019.
Polling station	Physical location where voters cast ballots on election day; assigned by the IEBC.
Sub-location	Official census-level administrative unit, below the ward.
Ward	Electoral sub-unit of a constituency; Nyali has five wards.

Acronyms and Prefixes

Acronym	Definition
AC	Acceptance Criterion
ASM	Assumption
AUD	Audit / Logging requirement
AUTH	Authentication requirement
AUTHZ	Authorisation requirement
BR	Business Rule
COMP	Compliance mapping
CON	Constraint
DEP	Dependency
DPA	Data Protection Act, 2019 (Kenya)
EDGE	Edge Case
ERR	Error Handling specification
FR	Functional Requirement
IN	Input specification
INT	Integration requirement
MVP	Minimum Viable Product
NFR	Non-Functional Requirement
OUT	Output specification
PERF	Performance target
POST	Postcondition

Acronym	Definition
PRE	Precondition
PWA	Progressive Web Application
RBAC	Role-Based Access Control
RLS	Row-Level Security (PostgreSQL feature)
SEC	Security requirement
TC	Test Case
UAT	User Acceptance Test
USAB	Usability requirement

2. Overall Description

2.1 Product Context

Alfayo Nelson Central Command is a greenfield platform with no system being replaced. It is operated by the campaign organisation through a fourteen-month development and operational cycle ending at the 9 August 2027 Kenya general election, after which the platform transitions to a constituency-relationship-maintenance mode for the candidate's parliamentary term (assuming election). The platform integrates with the WhatsApp Business API for field communications, a transactional email provider for notifications, an SMS gateway for election-day fallback, and a map tile provider for geographic visualisation. Field coordinators open the Field App on Android phones across the five wards of Nyali throughout each day; campaign leadership reviews Central Command on laptops in the campaign office during morning briefings and weekly strategy meetings; on election day, polling agents and ward coordinators use NyaliTrack continuously from polling-station opening at 06:00 through results transmission late in the evening.

2.2 User Classes and Characteristics

GUIDANCE	Every distinct user role is named with precise capabilities. Generic labels such as "admin" do not appear.	
User Class	Description	Key Permissions
Aspirant (Candidate)	Alfayo Nelson; sees strategic dashboards for briefings; rarely edits data	Read-only across all data except meetings owned by them
Campaign Manager	Senior operational lead; uses Central Command intensively daily	Full read/write across operations; cannot modify audit logs
Chief Strategist	Owns analysis and persuasion strategy; lives in the Pollings & Analysis tab	Full read; write on strategy and assessment fields
Constituency Coordinator	Cross-ward operational coordination; uses both Central Command and Field App	Full read across the constituency; write on assignments and activities
Media Head	Manages communications and meeting visibility	Read on activities and meetings; write on event communications
Comms Head	Operates the WhatsApp comms layer and broadcast lists	Read on people and activities; write on broadcasts
Patron / CEO	Senior oversight role; receives meeting notifications	Read-only across the platform
Ward Coordinator	Owns one ward; manages assistant coordinators, polling-station leads, agents, and canvassers	Read/write within their assigned ward only

Assistant Ward Coordinator	Deputy to Ward Coordinator; same scope, slightly reduced administrative rights	Read/write within their assigned ward; cannot remove people
Polling Station Lead	Owns operations at a single polling station; intensive on election day	Read/write on their station and the immediate village area
Polling Agent	Stationed at a polling station on election day; submits reports	Submit reports for their station; read assignment and event data
Canvasser	Door-to-door volunteer; receives daily target lists	Submit contact outcomes; read their assigned list only
Influence Liaison	Manages relationships with community leaders and sites within their ward	Read/write on community leaders and sites in their ward
Tech Lead	System administrator from Developers Mania	Full administrative access; subject to elevated audit logging
Finance Lead	Budget and procurement oversight	Read on activities; no access to voter or strategic data

2.3 Operating Environment

Client Environment

- Central Command web — Chrome 120+, Firefox 120+, Safari 17+, Edge 120+ on desktop or laptop
- Minimum resolution 1366×768; designed for 1920×1080
- Field App and NyaliTrack — Android 8.0 (Oreo) or later via Google Play Store; iOS 14 or later via Apple App Store
- Field App minimum device specification — 2 GB RAM, 16 GB storage, GPS, camera
- Connectivity — Central Command assumes broadband office connection; Field App and NyaliTrack must function on 3G and tolerate intermittent connectivity

Server / Cloud Environment

- Cloud platform — AWS Cape Town region (af-south-1) or a Nairobi-based provider with equivalent capability, to satisfy African data residency
- Runtime — Node.js 20 LTS on Ubuntu 22.04 LTS
- Database — PostgreSQL 15 or later with PostGIS 3.3 or later extension
- Cache and session store — Redis 7 or later
- File storage — Supabase Storage or AWS S3 with African region preference
- Web hosting — Vercel for Central Command; self-hosted or managed for the API
- Mobile build and deploy — Expo Application Services (EAS) Build and EAS Update

2.4 Constraints

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Each constraint is enforceable and traceable to a specific source

(regulatory, technical, or operational).

- CON-001 — All personal data of Kenyan citizens shall be stored on servers physically located in Africa, with Kenya preferred where feasible (DPA 2019 §50, data localisation principle).
- CON-002 — The campaign organisation shall be registered with the Office of the Data Protection Commissioner as a Data Controller before any voter-level data is ingested into the system (DPA 2019 §18–19).
- CON-003 — Technology stack is mandated: Node.js with Next.js for Central Command and the API layer, Expo with React Native for the mobile applications, PostgreSQL with PostGIS as the database, TypeScript throughout.
- CON-004 — All API communication shall use HTTPS with TLS 1.3 or later.
- CON-005 — Authentication shall require two-factor verification for all users at Ward Coordinator level and above.
- CON-006 — No personal data, voter information, or strategic intelligence shall be transmitted to any third-party AI or large-language-model service.
- CON-007 — Source code shall be private; no public repositories permitted.
- CON-008 — Mobile applications shall function offline for a minimum of twenty-four hours before requiring synchronisation.
- CON-009 — Field App and NyaliTrack shall never store the complete voter register on a device; voter lookups are on-demand and individually audit-logged.
- CON-010 — All processing of sensitive personal data shall have a documented lawful basis under the DPA; lawful bases will be confirmed by engaged legal counsel before production data ingestion.

2.5 Assumptions and Dependencies

Assumptions

- ASM-001 — IEBC polling station data for Nyali (boundaries, codes, voter counts, 2022 results) is available and reliable; the client has confirmed possession of this data.
- ASM-002 — The 9 August 2027 election date will not shift by more than six months; the system supports configurable election date but plans assume this date.
- ASM-003 — At least three ward coordinators will be available for user testing from Month 2 of development.
- ASM-004 — Legal counsel engaged by the campaign is available throughout development for DPA compliance review.
- ASM-005 — Ward coordinators have personal Android smartphones meeting the minimum specification; the campaign will not supply devices.
- ASM-006 — At least one team member has shipped a production React Native application before; if not, two weeks of upskilling are budgeted within Month 1.

Dependencies

- DEP-001 — WhatsApp Business API approval via a Business Solution Provider (360dialog or Africa's Talking); application must be submitted within Week 1 of development, approval typically four to eight weeks.
- DEP-002 — ODPC Data Controller registration — required before Month 5 (voter-data ingestion phase).
- DEP-003 — Africa's Talking SMS gateway account — required by Month 9 for election-day fallback testing.

- DEP-004 — Email transactional provider account (Resend or Postmark) — required by Month 2 for meeting-notification implementation.
- DEP-005 — Apple Developer Program account (\$99/year) — required by Month 1 for iOS build infrastructure.
- DEP-006 — Google Play Developer account (\$25 one-time) — required by Month 1.
- DEP-007 — Map tile provider account (MapLibre with OpenStreetMap, or Mapbox if budget permits) — required by Month 2.
- DEP-008 — Cloud hosting account in an African region — required by Month 1.
- DEP-009 — Third-party security audit firm engagement — required by Month 6, audit must complete before voter-level data is loaded in production.

3. Functional Requirements

3.1 Feature Prioritisation

This SRS uses the MoSCoW prioritisation scheme:

Priority	Definition
MUST	Non-negotiable for this release. Without it, the platform cannot be operated for the campaign cycle.
SHOULD	High value. Included after all MUSTs are complete and stable.
COULD	Nice to have. Deferrable to a later release without invalidating the current one.
WON'T	Explicitly out of scope for v1.0. Documented for clarity, not implementation.

Functional requirements are grouped by domain. Each domain uses a ten-step identifier block (FR-001+, FR-010+, FR-020+...) leaving gaps for future insertion without renumbering.

3.2 Identity and Access Management (FR-001 – FR-009)

This domain covers authentication, authorisation, session management, and role assignment across both Central Command and the mobile applications.

FR-001 — User Authentication

Priority	MUST
Source	Security baseline; Constraint CON-005
Dependencies	DEP-008 (cloud hosting)
User Story	As a campaign team member, I want to authenticate with my phone number and password so that I can access only the data and capabilities authorised for my role.
Description	The system shall authenticate users via phone number or email and password, with time-based one-time password (TOTP) two-factor verification required for users at Ward Coordinator level and above.

Preconditions

- PRE-001.1 — User account has been created and is active.
- PRE-001.2 — User role assignment is complete in the people table.

Inputs

- IN-001.1 — phone or email (string, required)
- IN-001.2 — password (string, minimum 12 characters, required)
- IN-001.3 — TOTP code (six-digit numeric, required for Ward Coordinator and above)

Postconditions

- POST-001.1 — A session token is issued and stored in the sessions table with device fingerprint and IP address.
- POST-001.2 — Authentication event is recorded in the audit log.
- POST-001.3 — User's last_active_at timestamp is updated.

Acceptance Criteria

- AC-001.1 — Valid credentials with correct TOTP (where required) → session issued, user redirected to role-appropriate home screen.
- AC-001.2 — Invalid password → request rejected with HTTP 401; failure counter incremented.
- AC-001.3 — Five consecutive failed attempts → account locked for 30 minutes; subsequent attempts return HTTP 423.
- AC-001.4 — Ward Coordinator and above without enrolled TOTP → forced to enrol on next login before access is granted.
- AC-001.5 — Successful login from a new device → email and WhatsApp notification sent to the user.

Business Rules

- BR-001.1 — Password complexity: minimum 12 characters, at least one uppercase, one lowercase, one number, one special character.
- BR-001.2 — Two-factor authentication is mandatory for the following roles: campaign_manager, chief_strategist, constituency_coordinator, ward_coordinator, assistant_ward_coordinator, polling_station_lead, tech_lead, admin.
- BR-001.3 — Session tokens expire after 24 hours on Central Command and 7 days on the mobile applications.

Error Handling

- ERR-001.1 — Invalid credentials → HTTP 401 with code AUTH_INVALID_CREDENTIALS; do not reveal whether the username or password was incorrect.
- ERR-001.2 — Account locked → HTTP 423 with code AUTH_ACCOUNT_LOCKED; include retry-after timestamp.
- ERR-001.3 — Missing required TOTP → HTTP 401 with code AUTH_2FA_REQUIRED.
- ERR-001.4 — Expired or revoked session token → HTTP 401 with code AUTH_SESSION_EXPIRED.

Audit Requirements

- AUD-001.1 — Every authentication event (success, failure, lockout, logout) is logged with user ID, IP address, device fingerprint, and timestamp.
- AUD-001.2 — Auth logs are retained for 7 years per DPA audit requirements; immutable append-only storage.

FR-002 — Role-Based Access Control

Priority	MUST
Source	Security baseline
Dependencies	FR-001
User Story	As a campaign manager, I want every user to see only the data their role authorises, so that voter and strategic information is

	protected even if individual accounts are compromised.
Description	The system shall enforce role-based access control at the database level using PostgreSQL row-level security policies, not in application code alone.

Acceptance Criteria

- AC-002.1 — A ward coordinator queries the people table → result set contains only people in their assigned ward.
- AC-002.2 — A canvasser queries the voters table → result set contains only voters on their assigned list for the current day.
- AC-002.3 — A polling agent attempts to read voters outside their station → query returns empty result set; attempt is audit-logged.
- AC-002.4 — Sensitive strategic notes on community leaders are visible only to candidate, campaign_manager, chief_strategist, and the owning team member.
- AC-002.5 — A direct SQL query bypassing the application layer still returns only authorised rows.

Business Rules

- BR-002.1 — RLS policies are defined per table and per role.
- BR-002.2 — The roles allowed to read sensitive_notes on community_leaders are: candidate, campaign_manager, chief_strategist, and the leader's owned_by_person.

Audit Requirements

- AUD-002.1 — Every read of community_leaders contact details and any voter record is logged with user, timestamp, IP, and queried record IDs.

FR-003 — Session Management

Priority	MUST
User Story	As an administrator, I want to revoke any active session immediately, so that compromised devices can be cut off.
Description	The system shall allow administrators to revoke any active session, immediately invalidating its token across all clients.

Acceptance Criteria

- AC-003.1 — Administrator revokes a session in the admin panel → subsequent API calls with that token receive HTTP 401 within 60 seconds.
- AC-003.2 — A user whose session is revoked receives a clear message on next interaction.
- AC-003.3 — Revocation event is recorded in the audit log with administrator ID and reason.

FR-004 — Biometric Mobile Unlock (Field App and Nyalitrack)

Priority	SHOULD
User Story	As a ward coordinator using the Field App many times daily, I want to unlock the app with fingerprint or Face ID after initial password login, so that I can work faster without compromising security.

Description	The mobile applications shall support biometric unlock as an alternative to password entry after the user has authenticated at least once.
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Acceptance Criteria

- AC-004.1 — User enables biometric unlock in settings → next app open prompts for fingerprint or Face ID.
- AC-004.2 — Three consecutive biometric failures → app falls back to password entry.
- AC-004.3 — Biometric unlock is disabled automatically if the device biometric set changes (e.g. user adds a fingerprint).

3.3 Geographic Intelligence (FR-010 – FR-019)

This domain covers the spatial spine of the platform: constituency, wards, sub-locations, villages, and polling stations.

FR-010 — Geographic Hierarchy Storage and Retrieval

Priority	MUST
Source	Product Owner brief; foundational for all spatial features
Dependencies	ASM-001 (IEBC data available)
User Story	As a campaign manager, I want every village in Nyali represented in the system with its proper ward and sub-location, so that activities and intelligence can be aggregated at any level.
Description	The system shall store the geographic hierarchy of Nyali Constituency (constituency → ward → sub-location → village → polling station) with PostGIS geometry for each level, enabling spatial queries across the hierarchy.

Acceptance Criteria

- AC-010.1 — Database is seeded with one constituency (Nyali), five wards (Kadzandani, Kongowea, Mkomani, Frere Town, Ziwa La Ng'ombe), all official sub-locations, and approximately 100 polling stations from IEBC data.
- AC-010.2 — Each entity has valid PostGIS geometry where boundary data is available, or a centroid Point where only location is known.
- AC-010.3 — Spatial query "all villages within ward X" returns the correct set.
- AC-010.4 — Spatial query "polling stations within 1 km of point P" returns ordered results with distance.

Business Rules

- BR-010.1 — Village/mtaa boundaries are approximate for informal settlements; the system stores centroid and optional polygon, and surfaces uncertainty in the UI.
- BR-010.2 — Polling station IEBC codes are unique and immutable; they are the canonical join key with IEBC data.

FR-011 — Village and Mtaa Management

Priority	MUST
User Story	As a ward coordinator, I want to add new villages and edit existing village records, so that the geographic data reflects ground reality.
Description	The system shall allow ward coordinators and above to create, edit, and delete village records within their authorised ward, capturing name, aliases, centroid, optional boundary, population estimate, and assigned coordinator.

Acceptance Criteria

- AC-011.1 — Ward coordinator adds a new village from the Field App with GPS auto-capture → record appears in their ward within 30 seconds of sync.

- AC-011.2 — Edits to a village record are versioned in the audit log.
- AC-011.3 — Deletion is soft (record marked inactive, not removed); only campaign manager and above may delete.
- AC-011.4 — Village name must be unique within a ward; duplicates trigger a merge prompt.

FR-012 — Constituency Map Visualisation

Priority	MUST
User Story	As the candidate, I want to see Nyali on the Central Command homepage as a clickable map with the five wards in distinct colours, so that I can navigate to any ward in one click.
Description	The system shall render the Nyali constituency map on the Central Command homepage with each of the five wards in a distinct colour, with each ward clickable to navigate to its detail page.

Acceptance Criteria

- AC-012.1 — Homepage map loads in under 2 seconds on a 10 Mbps connection.
- AC-012.2 — Hovering over a ward displays its name and a one-line summary (registered voters, last activity timestamp).
- AC-012.3 — Clicking a ward navigates to the WARDS tab for that ward.
- AC-012.4 — Map is rendered with MapLibre GL JS using OpenStreetMap tiles.

3.4 Community Intelligence (FR-020 – FR-029)

This domain is the strategic core of the platform: community leaders, community sites, and influence networks across Nyali.

FR-020 — Community Leader Directory

Priority	MUST
Source	Product Owner brief; identified as highest-value data asset
Dependencies	FR-002 (RBAC), FR-010 (geographic hierarchy)
User Story	As a ward coordinator, I want every imam, pastor, boda chairman, mama kiongozi, and elder in my ward recorded with their role, location, and our relationship status, so that the network survives when individuals leave the team.
Description	The system shall maintain a structured directory of community leaders separate from campaign team members, capturing identity, role, affiliation, geographic anchor, influence reach, political lean, relationship temperature, ownership, and contact history.

Inputs

- IN-020.1 — full_name (string, required)
- IN-020.2 — phone, whatsapp_phone, email (strings, optional)
- IN-020.3 — role_title (string, required) — Imam, Pastor, Boda Chairman, Mama Kiongozi, Chief, Elder, etc.
- IN-020.4 — affiliated_site (reference to community_sites, optional)
- IN-020.5 — village and ward (references, required)
- IN-020.6 — influence_reach (enum: small, medium, large, unknown)
- IN-020.7 — political_lean (enum: supportive, leaning_supportive, neutral, leaning_opposition, opposition, unknown)
- IN-020.8 — relationship_temperature (enum: warm, cool, cold, hostile, not_approached)

Acceptance Criteria

- AC-020.1 — Ward coordinator adds a new community leader from the Field App with GPS auto-capture of location.
- AC-020.2 — New entries from canvassers and polling agents are queued for ward coordinator review before entering the primary directory.
- AC-020.3 — Duplicate detection: if a leader with the same name in the same ward already exists, the system prompts the submitter to confirm or merge.
- AC-020.4 — Searching the directory returns results filtered by the submitter's role-based visibility.
- AC-020.5 — Sensitive strategic notes are visible only to candidate, campaign manager, chief strategist, and the owning team member.

Business Rules

- BR-020.1 — Every community leader has an owned_by_person assigned; ownership transfers are logged.

- BR-020.2 — A leader's political_lean and relationship_temperature may be edited only by the owner or campaign manager and above.

Audit Requirements

- AUD-020.1 — Every read of contact details (phone, whatsapp_phone, email) is logged with user, timestamp, and IP.
- AUD-020.2 — Every modification of political_lean, relationship_temperature, or sensitive_notes is logged with before and after values.

FR-021 — Community Sites Directory

Priority	MUST
User Story	As an influence liaison, I want every mosque, church, school, market, boda stage, and chama in my ward recorded with location, size, schedule, and key contact, so that the candidate can visit the right places at the right times.
Description	The system shall maintain a directory of community sites with type, name, location, ward, estimated size, meeting schedule, key contact, political climate assessment, and visit history.

Acceptance Criteria

- AC-021.1 — Site types include: church, mosque, madrasa, school (primary, secondary, other), market, shopping_center, boda_stage, matatu_stage, chama, sacco, self_help_group, community_hall, social_hall, sports_club, youth_center, health_facility, government_office, other.
- AC-021.2 — Field App allows adding a site with three taps and GPS capture.
- AC-021.3 — Sites are linked to a village and optionally to a key contact in the community_leaders directory.
- AC-021.4 — Each site displays its last_visited_at and the activity log of past visits.

FR-022 — Boda Boda Stage Aggregate Tracking

Priority	MUST
User Story	As a chief strategist, I want boda boda stages tracked at the stage level rather than the individual rider level, so that we capture political influence without profiling individual citizens.
Description	The system shall track boda boda stages as community_sites with chairman as key contact and estimated rider count, but shall not store records of individual riders.

Acceptance Criteria

- AC-022.1 — A boda stage record includes location, chairman (linked to community_leaders), estimated rider count, and political climate.
- AC-022.2 — The system rejects attempts to create individual rider records; only the stage chairman and assistant chairman are stored.

3.5 Issue Management (FR-030 – FR-039)

Village-level issue tracking generates the talking-points data the candidate needs before every baraza and door-to-door sweep.

FR-030 — Village Issue Capture

Priority	MUST
Source	Product Owner brief
Dependencies	FR-010 (villages)
User Story	As a ward coordinator who just heard residents complain about a burst sewer at a baraza, I want to record the issue against the village in 30 seconds, so that the candidate sees it before the next visit.
Description	The system shall allow authorised users to record village-level issues with category, title, description, severity, estimated voters affected, status, current administration responsiveness, the candidate's position, and verification status.

Inputs

- IN-030.1 — village_id (reference, required)
- IN-030.2 — category (enum from controlled vocabulary, required)
- IN-030.3 — title (string, required, max 120 characters)
- IN-030.4 — description (text, optional)
- IN-030.5 — severity (enum: minor, moderate, serious, critical, required)
- IN-030.6 — affects_estimated_voters (integer, optional)

Acceptance Criteria

- AC-030.1 — Issue categories are constrained to the controlled vocabulary defined in BR-030.1.
- AC-030.2 — Issues are displayed on village detail pages and rolled up to ward pages with severity counts.
- AC-030.3 — Issues not verified within 90 days are flagged "stale".
- AC-030.4 — Issues can be filtered by ward, village, category, severity, and status.

Business Rules

- BR-030.1 — Issue categories: water, sanitation, garbage, drainage, roads, street_lighting, electricity, security, drugs_substance_abuse, gbv, youth_unemployment, education, healthcare, land_disputes, housing, business_permits, agriculture, fishing, environmental, corruption, service_delivery, other.
- BR-030.2 — Issues with severity "critical" trigger a notification to the campaign manager.

3.6 Activity and Event Management (FR-040 – FR-049)

FR-040 — Campaign Activity Lifecycle

Priority	MUST
Source	Product Owner brief
Dependencies	None
User Story	As a campaign manager, I want every rally, baraza, mosque visit, and door-to-door sweep planned, executed, and outcome-tracked in one system, so that we know what worked and what did not.
Description	The system shall support the full lifecycle of campaign activities — planning, confirmation, execution, attendance recording, outcome notes, and follow-up tracking — across the controlled list of activity types.

Inputs

- IN-040.1 — title, type, scheduled_at, location, ward, village (required)
- IN-040.2 — expected_attendance (integer, optional)
- IN-040.3 — owner (person reference, required)

Postconditions

- POST-040.1 — Activity record is created and visible in role-appropriate lists.
- POST-040.2 — If status is "confirmed", invitees receive notifications per FR-060.

Acceptance Criteria

- AC-040.1 — Activity types include: rally, baraza, community_meeting, town_hall, mosque_visit, church_visit, madrasa_visit, boda_stage_stop, market_visit, chama_meeting, door_to_door, youth_event, women_event, harambee, condolence_visit, wedding_attendance, courtesy_call, media_engagement, launch_event, internal_strategy, training, other.
- AC-040.2 — After an activity is completed, the owner can record actual_attendance, candidate_attended flag, outcome notes, follow_up_required flag, and the list of community leaders who attended.
- AC-040.3 — Activities can be filtered in Central Command by ward, village, type, owner, status, and date range.
- AC-040.4 — Photos may be attached to an activity (optional, with consent metadata captured per FR-080).

3.7 Visit Logging and Coverage Heatmaps (FR-050 – FR-059)

FR-050 — Three-Tap Visit Logging

Priority	MUST
Source	Product Owner brief — usability is critical for field adoption
Dependencies	FR-001, FR-010
User Story	As a canvasser walking through a Kongowea estate, I want to log a visit in three taps without breaking my conversation rhythm, so that the data actually gets captured.
Description	The Field App shall allow a user to log a visit in three taps or fewer from app open, capturing visiting person, GPS location, auto-suggested village, purpose, and optional engagement count.

Acceptance Criteria

- AC-050.1 — From the Field App home screen, three taps reach a submitted visit record: (1) "Log Visit" button, (2) purpose selection, (3) submit.
- AC-050.2 — GPS is auto-captured and the nearest village is auto-suggested; user can override.
- AC-050.3 — The flow works fully offline; submissions queue in the local SQLite database and sync when connectivity returns.
- AC-050.4 — A submitted offline visit appears in Central Command within 60 seconds of the device regaining connectivity.

Business Rules

- BR-050.1 — Visits are append-only event records; they cannot be edited after submission, only annotated.

FR-051 — Coverage Heatmap (Heatmap A)

Priority	MUST
Source	Product Owner brief
Dependencies	FR-050
User Story	As a campaign manager planning next week, I want to see at a glance which villages have not been visited recently, so that I can deploy people to the gaps.
Description	The Central Command shall render a Coverage heatmap over the Nyali constituency map using a hex-grid overlay (300–500m hexes), with colour intensity weighted by recency of visits within the selected time window.

Acceptance Criteria

- AC-051.1 — Heatmap renders for the full constituency in under 3 seconds.
- AC-051.2 — Time window is configurable: last 7, 30, 90, or 365 days.
- AC-051.3 — Filtering by person, activity type, or ward updates the heatmap.

- AC-051.4 — Recency weighting: visits within 7 days have weight 1.0; weight decays linearly to 0.0 at the end of the window.

FR-052 — Influence Density Heatmap (Heatmap B)

Priority	MUST
User Story	As a chief strategist, I want to see where we have the most warm-contact community leaders, so that I know where our human infrastructure is strong or thin.
Description	The system shall render an Influence Density heatmap showing the count and reach of warm-contact community leaders per geographic area, over the same hex-grid as Heatmap A.

Acceptance Criteria

- AC-052.1 — Influence weight = sum of (relationship_temperature score × influence_reach score) for leaders in each hex.
- AC-052.2 — Heatmap is filterable by leader role type (imams only, boda chairmen only, etc.).

FR-053 — Persuasion Opportunity Heatmap (Heatmap C)

Priority	MUST
User Story	As a campaign manager, I want a computed heatmap of where the next week's effort should focus, combining historical margin, voter density, coverage gaps, and issue alignment.
Description	The system shall render a Persuasion Opportunity heatmap computed from 2022 vote margin (closer = higher), voter density, coverage gap (less recent contact = higher), and issue alignment (villages whose top issues match the candidate platform = higher).

Acceptance Criteria

- AC-053.1 — The weighting formula is documented in administrator settings and adjustable.
- AC-053.2 — Hovering on a hex shows the contributing factors and the underlying numbers.

3.8 Meeting Coordination (FR-060 – FR-069)

FR-060 — Meeting Creation with Automated Notifications

Priority	MUST
Source	Product Owner brief
Dependencies	DEP-001 (WhatsApp), DEP-004 (email)
User Story	As a constituency coordinator, I want to schedule a meeting with elders in Kongowea and have the patron, CEO, and media head automatically notified by WhatsApp and email, so that I do not have to remember to forward to everyone.
Description	The system shall allow authorised users to create meetings with title, date, time, location, agenda, type, and invitee list. On creation and modification, the system shall automatically notify configured stakeholders (default: patron, CEO, media head) via both WhatsApp and email.

Acceptance Criteria

- AC-060.1 — Meeting types include: internal_strategy, community_baraza, stakeholder, condolence_visit, harambee, courtesy_call, media.
- AC-060.2 — On creation, WhatsApp and email notifications are dispatched to configured stakeholders within 60 seconds.
- AC-060.3 — Invitees receive WhatsApp reminders 24 hours and 1 hour before the scheduled time, configurable per meeting.
- AC-060.4 — If WhatsApp delivery fails, the system falls back to SMS via Africa's Talking.
- AC-060.5 — Past meetings without outcome notes after 48 hours are flagged to the meeting owner.

Business Rules

- BR-060.1 — The notification stakeholder list is configurable per organisation in administrator settings; it is not hardcoded to patron/CEO/media head.
- BR-060.2 — Meeting deletion sends a cancellation notification to all invitees and stakeholders.

Error Handling

- ERR-060.1 — WhatsApp API failure → log error, retry with exponential backoff up to 3 times, then fall back to SMS.
- ERR-060.2 — Both WhatsApp and SMS failure → alert the meeting owner via in-app notification and email.

3.9 Pollings and Analysis Dashboards (FR-070 – FR-079)

FR-070 — Electoral History Dashboard

Priority	MUST
User Story	As a chief strategist, I want to see results from the 2013, 2017, and 2022 elections at ward and polling-station level, so that I can build informed persuasion strategy.
Description	The Central Command shall display electoral history dashboards covering 2013, 2017, and 2022 results for presidential, parliamentary, gubernatorial, senatorial, and MCA contests, with ward-level and polling-station-level granularity.

Acceptance Criteria

- AC-070.1 — Turnout trends by ward across the three elections are visualised.
- AC-070.2 — Margin and competitiveness maps are available per ward.
- AC-070.3 — Results can be exported as PNG and CSV.

FR-071 — Demographics Dashboard

Priority	MUST
User Story	As a campaign manager, I want voter age, gender, and (carefully) ethnic composition by ward, so that the campaign can plan resonance-driven outreach.
Description	The system shall display demographics dashboards based on IEBC voter register figures and KNBS 2019 Census data, including registered voter counts, age and gender distribution, and ethnic composition where data permits.

Acceptance Criteria

- AC-071.1 — Ethnic composition is presented only at ward level or higher, never at individual or polling-station level.
- AC-071.2 — Data source and date are visible on each chart.

FR-072 — Current Race Intelligence

Priority	MUST
Source	Product Owner brief; sensitive feature requiring methodology transparency
Dependencies	None
User Story	As a campaign manager, I want a current-race section showing other declared candidates, their parties, and the reception they are receiving, with the methodology of each assessment clearly labelled.
Description	The Central Command shall display a Current Race Intelligence section covering declared opposing candidates, their party

	affiliations, reception assessment, MCA-level analysis per ward, and party popularity trends — with every assessment labelled by methodology (coordinator reports, commissioned poll, public statement, etc.).
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Acceptance Criteria

- AC-072.1 — Every reception assessment displays a methodology label and the assessment date.
- AC-072.2 — Source data for each assessment is recorded in the database and viewable by candidate, campaign_manager, and chief_strategist.
- AC-072.3 — Strategic notes on opposition candidates ("can be flipped", "weak in ward X") are stored in a restricted-access field visible only to candidate, campaign_manager, and chief_strategist.

Business Rules

- BR-072.1 — Assessments older than 60 days are flagged "stale" and require re-verification.
- BR-072.2 — The platform shall not aggregate restricted strategic notes into exported reports.

3.10 Team Management (FR-080 – FR-089)

FR-080 — Team Directory and Organisation Chart

Priority	MUST
User Story	As the candidate, I want to see the whole team in a clear hierarchy with photos, so that I and visitors can put faces to roles.
Description	The system shall display the team in two views: a hierarchical list (Executive & Technical Support, then per-ward sections) and a visual org chart, with each person card showing photo, name, role, phone, WhatsApp, email, ward, and activity status.

Acceptance Criteria

- AC-080.1 — Activity status indicator: "Active" if used the app in the last 7 days; "Inactive" otherwise.
- AC-080.2 — Clicking a person reveals their assignment history, recent visits, and reports submitted.
- AC-080.3 — Reverse lookup: "Who covers village X?" returns the assigned coordinator and active canvassers.
- AC-080.4 — Photos are optional; absent photos display a clean initials placeholder.

3.11 Homepage, Navigation, and Countdown (FR-090 – FR-099)

FR-090 — Election Countdown Widget

Priority	MUST
Source	Product Owner brief
Dependencies	None
User Story	As any user of Central Command, I want to see the time remaining until election day at all times, so that the urgency is felt by everyone who opens the app.
Description	The Central Command shall display a persistent countdown widget in the top navigation bar showing months, days, hours, and minutes until the configured election date (default: 9 August 2027).

Acceptance Criteria

- AC-090.1 — Countdown is visible on every page of Central Command.
- AC-090.2 — The election date is configurable by administrators.
- AC-090.3 — When the date passes, the widget displays "Election day" and then a count of days since.

FR-091 — Central Command Homepage Composition

Priority	MUST
User Story	As a senior user opening Central Command, I want the homepage to anchor on the constituency map with key indicators flanking it, so that I can navigate or sense the state of the campaign immediately.
Description	The Central Command homepage shall display, in vertical order: top navigation with countdown; main banner with countdown; clickable constituency map with the five wards in distinct colours; ward summary carousel; pollings and analysis preview strip; team leads section; upcoming and pending meetings.

Acceptance Criteria

- AC-091.1 — The constituency map is the visual anchor and occupies at least 40 percent of viewport height on a 1920×1080 display.
- AC-091.2 — Ward summary carousel shows registered voters, last activity, top issue, coverage percentage.
- AC-091.3 — Upcoming meetings section shows the next five meetings the user is invited to or owns.

FR-092 — NyaliTrack Navigation Button

Priority	MUST
User Story	As an authorised user, I want a clear button labelled

	NYALITRACK that opens the election-day operations app, so that on D-Day I switch contexts cleanly.
Description	The Central Command navigation shall include a NYALITRACK button that opens the election-day operations application in a new context.

Acceptance Criteria

- AC-092.1 — Before election day, the button opens an informational page describing what NyaliTrack will do on D-Day.
- AC-092.2 — From 7 days before election day, the button opens the NyaliTrack staging environment for dress rehearsals.
- AC-092.3 — On election day, the button opens the production NyaliTrack interface.

3.12 Field App Specific Requirements (FR-100 – FR-109)

FR-100 — Offline-First Operation

Priority	MUST
Source	Constraint CON-008; field reality in Nyali
Dependencies	FR-001, FR-050
User Story	As a canvasser working in an area with weak 3G, I want the Field App to work fully offline and sync when connectivity returns, so that no data is lost.
Description	The Field App shall function offline for a minimum of 24 hours, storing all pending submissions in a local SQLite database, and shall automatically synchronise pending submissions when connectivity is restored using background fetch tasks where supported by the operating system.

Acceptance Criteria

- AC-100.1 — With airplane mode enabled, the user can log visits, add community leaders, add sites, and submit reports for 24 hours without errors.
- AC-100.2 — On reconnection, pending submissions are synced within 5 minutes, retrying with exponential backoff on individual failures.
- AC-100.3 — Submission unique IDs are generated client-side to prevent duplicate inserts on retry.
- AC-100.4 — Conflict resolution on mutable records uses last-write-wins; append-only event records (visits, reports) never conflict.

Error Handling

- ERR-100.1 — Sync failure after 5 retries → submission remains in the local queue; user is notified via in-app banner.
- ERR-100.2 — Local storage near capacity → user is warned at 80 percent and prompted to sync.

FR-101 — Role-Aware Mobile Home Screen

Priority	MUST
User Story	As a Field App user, I want the home screen to match my role — short and quick for agents and canvassers, more complete for ward coordinators — so that I can do my job without scrolling through irrelevant options.
Description	The Field App shall present a role-aware home screen: polling agents and canvassers see today's tasks and quick report buttons; ward coordinators additionally see ward summary, team status indicators, and assignment tools.

Acceptance Criteria

- AC-101.1 — Agent home shows: today's assignment card, "Log Visit" button, "Submit Report" button, messages.

- AC-101.2 — Ward coordinator home shows: ward summary (active agents, last 24h reports, flagged issues), my team list, assignment tools, messages, plus the agent-level quick buttons.
- AC-101.3 — Home screen rendering does not require a network call after initial sync.

FR-102 — Push Notifications

Priority	MUST
User Story	As a Field App user, I want push notifications for meeting reminders, urgent broadcasts, and (on election day) incident escalations, so that I am reachable without keeping the app open.
Description	The Field App shall deliver push notifications via Expo Notifications for meeting reminders, assignment alerts, urgent broadcasts, and election-day incident escalations.

Acceptance Criteria

- AC-102.1 — Notifications are scoped to the user's role and assigned ward; a polling agent does not receive notifications about other wards.
- AC-102.2 — Critical notifications (election day incidents) bypass Do Not Disturb settings on Android where the OS permits.

3.13 Search and Export (FR-110 – FR-119)

FR-110 — Global Search

Priority	MUST
User Story	As any user, I want a global search across the platform so that I can find a person, village, leader, site, issue, meeting, or activity quickly.
Description	The system shall provide global search across people, villages, community leaders, community sites, issues, meetings, and activities, with results filtered by the user's role-based visibility.

Acceptance Criteria

- AC-110.1 — Search returns ranked results within 800 ms at the 90th percentile.
- AC-110.2 — Partial-string matching and common Swahili-English variants are supported (e.g. "msikiti" matches "mosque").
- AC-110.3 — Results are grouped by entity type with counts.

FR-111 — Data Export

Priority	SHOULD
User Story	As a campaign manager, I want to export ward briefing packs as PDF and contact lists as CSV, so that the candidate can prepare for a baraza on the road.
Description	The system shall support PDF export of ward briefing packs, village briefing packs, and meeting agendas; and CSV export of visit logs, activity records, community leader directories, and issue registers.

Acceptance Criteria

- AC-111.1 — Every export is logged in the audit trail with user, timestamp, and scope.
- AC-111.2 — Exports are scoped by the user's role-based visibility.
- AC-111.3 — PDF exports include the export timestamp and a "Confidential" watermark.

3.14 NyaliTrack Election-Day Operations (FR-120 – FR-129)

NyaliTrack is outlined here at functional-domain level. A full election-day SRS (SRS-ALFAYO-002) will be produced in Month 6 once campaign-mode data structures are stable.

FR-120 — Polling Station Status Board

Priority	MUST
User Story	As the war room on election day, I want a single board showing the status of every polling station — agent check-in, turnout pace, incident flags — so that we deploy resources to stations that need attention.
Description	NyaliTrack shall display a real-time status board for every polling station in Nyali, showing agent check-in status, turnout pace vs. target, last report timestamp, and any flagged incidents.

Acceptance Criteria

- AC-120.1 — Status board updates within 60 seconds of an agent submitting a report.
- AC-120.2 — Stations with no agent check-in by 06:30 are flagged red.
- AC-120.3 — Stations whose turnout pace falls below target by more than 20 percent are flagged amber.

FR-121 — Structured Incident Reporting

Priority	MUST
User Story	As a polling agent, I want to submit an incident report in three taps with structured categories, so that the war room can triage without reading paragraphs.
Description	NyaliTrack shall support structured incident reports with categories: voter intimidation, agent obstruction, ballot issue, materials shortage, violence, dispute, technical failure, other; each with severity, free-text description, optional photo, and GPS.

Acceptance Criteria

- AC-121.1 — Incident submission works offline with priority queue (incidents sync before routine reports).
- AC-121.2 — Critical-severity incidents trigger push notifications to ward coordinator and constituency coordinator within 30 seconds of network reach.

FR-122 — SMS Fallback for D-Day

Priority	MUST
User Story	As a polling agent on a degraded network, I want to submit turnout numbers via SMS short code as a fallback, so that data still reaches the war room.
Description	NyaliTrack shall support structured SMS reporting via an Africa's Talking short code as a fallback when data network is unavailable.

Acceptance Criteria

- AC-122.1 — SMS format is documented and printed on a card distributed to every agent before election day.
- AC-122.2 — Inbound SMS are parsed and inserted into the same reports table as app submissions, marked with source=sms.

3.15 Committed Supporter Network (FR-130 – FR-139)

This domain manages the list of voters who have indicated commitment to support the candidate, typically arising from their participation as beneficiaries of community programs (bursaries, harambees, water projects, women's groups, youth programs, medical support, and similar initiatives). This is the most sensitive personal data the platform holds and is governed by stricter access controls, audit obligations, and consent requirements than any other domain.

SENSITIVITY NOTE

Data in this domain is classified as Restricted. Records link named voters to participation in community programs, which may reveal economic circumstances, health information, and inferred political affiliation. The Kenya Data Protection Act, 2019 §44–45 treats such linkages as sensitive personal data requiring elevated lawful basis, explicit consent, and additional safeguards. No record may be created in this domain without a documented consent capture event per BR-130.1.

FR-130 — Committed Supporter Registration

Priority	MUST
Source	Product Owner brief; identified as high-value campaign asset
Dependencies	FR-002 (RBAC), FR-010 (geographic hierarchy), DEP-002 (ODPC registration), COMP-010 (lawful basis)
User Story	As a ward coordinator who has confirmed a beneficiary of one of the candidate's community programs intends to vote for Alfayo Nelson, I want to register them as a committed supporter with the program through which we built the relationship, so that the campaign can plan accurate turnout operations and the relationship can be honoured after the election.
Description	The system shall maintain a registry of voters who have indicated commitment to support the candidate, capturing identity, polling station, originating community program, commitment tier, registering team member, consent capture metadata, and last-verified date. Records in this domain are subject to elevated access controls, audit logging, and export restrictions.

Preconditions

- PRE-130.1 — Voter exists in the voters table or is captured at the same time with verified IEBC polling-station assignment.
- PRE-130.2 — A community_program record exists or is created at the same time, identifying the program through which the supporter was engaged.
- PRE-130.3 — A consent capture event has been recorded (verbal at minimum, with the registering team member as witness).
- PRE-130.4 — The voter is not flagged "do_not_contact" in the system.

Inputs

- IN-130.1 — voter_id (UUID, required)
- IN-130.2 — community_program_id (UUID, required)

- IN-130.3 — `commitment_tier` (enum: `strong_commit`, `likely`, `probable`, `unverified`, required)
- IN-130.4 — `registering_person_id` (UUID, required)
- IN-130.5 — `consent_capture_method` (enum: `verbal_witnessed`, `written_signed`, `sms_confirmed`, `in_person_app`, required)
- IN-130.6 — `consent_captured_at` (timestamp, required)
- IN-130.7 — `notes` (text, optional)

Postconditions

- POST-130.1 — Supporter record is created in the `committed_supporters` table with restricted-access flag set.
- POST-130.2 — Consent capture event is recorded in the `consent_log` table (append-only).
- POST-130.3 — Audit log records the creation event with registering person, IP, and timestamp.
- POST-130.4 — The supporter's `commitment_tier` is recalculated by the system if multiple registrations exist (highest tier wins, lowest age wins as tiebreaker).

Acceptance Criteria

- AC-130.1 — Ward coordinator registers a supporter from the Field App after confirming verbal consent → record is created within 60 seconds of sync.
- AC-130.2 — Registration without a documented consent capture method is rejected with a clear error.
- AC-130.3 — Duplicate registration (same voter, same program) prompts merge or update rather than creating a second record.
- AC-130.4 — Records are visible only to candidate, `campaign_manager`, `chief_strategist`, `constituency_coordinator`, and the assigning `ward_coordinator` (not other ward coordinators).
- AC-130.5 — Every read of a supporter record is logged in the `audit_log` with reader identity, timestamp, and IP.
- AC-130.6 — Bulk exports do not include the `committed_supporters` table by default; export requires `campaign_manager` approval recorded in the audit log.

Business Rules

- BR-130.1 — A documented consent capture is mandatory before any supporter record is created. Consent must include the supporter's explicit awareness that they are being recorded as a committed supporter of the campaign.
- BR-130.2 — Commitment tier values: `strong_commit` (explicit verbal commitment + ongoing relationship), `likely` (positive engagement, no explicit commitment), `probable` (program beneficiary without explicit political conversation), `unverified` (relationship inferred but not confirmed).
- BR-130.3 — The platform shall not infer commitment tier automatically from program participation; tier is set explicitly by the registering coordinator based on direct knowledge.
- BR-130.4 — A supporter may withdraw consent at any time. Upon withdrawal, the record is soft-deleted (marked `withdrawn=true` with timestamp); reads of withdrawn records are restricted to `campaign_manager` and above for audit purposes only.
- BR-130.5 — The term "guaranteed" is not used anywhere in the user interface or in exports; "committed supporter" with explicit tier is the canonical language.
- BR-130.6 — The `committed_supporters` table is never used for targeted messaging without separate, message-specific consent.

Error Handling

- ERR-130.1 — Attempt to register a supporter without consent metadata → HTTP 400 with code SUPPORTER_CONSENT_REQUIRED.
- ERR-130.2 — Attempt to read a supporter record by an unauthorised role → HTTP 404 (not 403, to avoid confirming existence of the record); attempt is audit-logged for review.
- ERR-130.3 — Attempt to bulk export without campaign_manager approval → HTTP 403 with code SUPPORTER_EXPORT_DENIED; approval workflow is initiated automatically.

Audit Requirements

- AUD-130.1 — Every create, read, update, withdraw, and export-attempt on committed_supporters and consent_log is recorded in the append-only audit_log with full context.
- AUD-130.2 — Audit log entries for this domain are retained for 7 years per DPA audit obligations and the campaign's elevated retention policy for sensitive data.
- AUD-130.3 — A monthly audit report listing every read of committed_supporters by user and volume is generated and reviewed by the campaign_manager and the Data Protection Officer.

FR-131 — Community Program Registry

Priority	MUST
Source	Required by FR-130 (programs are the source of supporter relationships)
Dependencies	FR-002, FR-010
User Story	As a campaign manager, I want every community program the candidate runs or supports recorded with its scope, beneficiary count estimate, and lead coordinator, so that supporter records can be traced to the relationship that built them.
Description	The system shall maintain a registry of community programs (bursaries, harambees, water projects, women's groups, youth programs, medical support, and similar initiatives) with name, type, ward and village scope, lead coordinator, active dates, estimated beneficiary count, and a brief description.

Acceptance Criteria

- AC-131.1 — Program types include: bursary, harambee, water_project, sanitation_project, women_group_support, youth_program, medical_support, sports_sponsorship, religious_donation, agricultural_input, business_capital, other.
- AC-131.2 — Programs are scoped to one or more wards and optionally one or more villages.
- AC-131.3 — A program may have multiple committed_supporters linked to it; counts roll up to program-level dashboards.
- AC-131.4 — Program records are visible to all team members at Ward Coordinator level and above; individual supporter records linked to a program remain restricted per FR-130.

Business Rules

- BR-131.1 — Programs explicitly involving cash transfers to voters in exchange for votes are prohibited by Kenyan election law and shall not be recorded in this system. The platform is not a tool for documenting electoral offences.
- BR-131.2 — Each program has a designated lead_coordinator who is accountable for the beneficiary list and ongoing consent.

Audit Requirements

- AUD-131.1 — Program creation, modification, and deletion are audit-logged with full context.

FR-132 — Committed Supporter Dashboard

Priority	MUST
User Story	As a campaign manager, I want a dashboard view of committed supporters by ward, polling station, commitment tier, and originating program, so that we can plan turnout operations accurately.
Description	The Central Command shall provide a Committed Supporter dashboard showing counts by ward, polling station, commitment tier, and originating program. Individual supporter records are accessible only to users authorised under FR-130; the dashboard presents aggregated counts only.

Acceptance Criteria

- AC-132.1 — Aggregated counts are visible to all users at Ward Coordinator level and above for their authorised ward; constituency-wide counts are visible only to candidate, campaign_manager, chief_strategist, and constituency_coordinator.
- AC-132.2 — Drill-through to individual records is gated by FR-130 access rules.
- AC-132.3 — Dashboard never displays voter names or contact details; identifiers and aggregates only.
- AC-132.4 — Dashboard exports as image only (no underlying data), and the export is audit-logged.

FR-133 — Commitment Tier Re-Verification

Priority	MUST
User Story	As a ward coordinator, I want the system to flag committed supporters whose tier has not been verified in 90 days, so that we keep the data current and honest.
Description	The system shall flag committed supporter records as "stale" if commitment_tier has not been verified by direct conversation with the supporter within the last 90 days. Stale records are excluded from turnout planning aggregates until re-verified.

Acceptance Criteria

- AC-133.1 — Stale records appear on a ward coordinator's home screen in the Field App as a re-verification queue.
- AC-133.2 — Re-verification is recorded as a new event in the consent_log with the verifier, method, and timestamp.

- AC-133.3 — Stale records older than 180 days require campaign_manager approval to retain or are automatically downgraded to commitment_tier=unverified.

FR-134 — Consent Withdrawal and Right to Erasure

Priority	MUST
Source	DPA §26 (data subject rights); ethical baseline
Dependencies	FR-130, FR-131
User Story	As a person previously registered as a committed supporter, I want to withdraw my consent and have my record removed or anonymised, so that my rights under the Kenya Data Protection Act are respected.
Description	The system shall support consent withdrawal for any committed supporter on request, with a documented procedure for handling the request within 30 days as required by the Kenya Data Protection Act, 2019 §26.

Acceptance Criteria

- AC-134.1 — A documented withdrawal procedure is published, accessible to data subjects via the campaign's Data Protection Officer contact.
- AC-134.2 — Upon receiving a withdrawal request, the system supports two actions: soft-delete (record marked withdrawn=true, retained for audit only) or hard-delete (record anonymised with personal data fields nulled, retained only as a count).
- AC-134.3 — Withdrawal is processed within 30 days of request; the requester receives written confirmation.
- AC-134.4 — Withdrawal events are logged in the consent_log with the requester, method of request, processing time, and outcome.

Business Rules

- BR-134.1 — Hard-deletion is the default for withdrawal requests unless the campaign's Data Protection Officer determines audit retention is legally required.
- BR-134.2 — A withdrawn supporter cannot be re-registered without a fresh consent capture event.

Audit Requirements

- AUD-134.1 — Every withdrawal request is logged with full context including the original consent capture event ID.
- AUD-134.2 — Anonymisation operations are logged with the user who performed them and the fields that were nulled.

4. Non-Functional Requirements

CRITICAL RULE

Every NFR has all five fields, every time. An NFR without a Measurable Target and a defined Acceptance Test is not an NFR — it is a wish.

4.1 Performance NFRs

NFR-001 — Central Command Page Load

Priority	MUST
Category	Performance
Description	Central Command pages shall load within 2 seconds under typical office broadband conditions.
Measurable Target	P95 page load time $\leq$ 2 seconds on a 10 Mbps connection, measured at 200 concurrent users.
Acceptance Test	Synthetic load test using k6 and Playwright over a 5-minute sustained run; P95 read from k6 summary; results archived in /docs/perf/.

NFR-002 — Field App First Load on 3G

Priority	MUST
Category	Performance
Description	The Field App shall complete initial load within 5 seconds on a 3G connection; cached subsequent loads within 1 second.
Measurable Target	Initial install + first-open + login completes in $\leq$ 5 s on simulated 3G (1 Mbps, 300 ms RTT). Cached subsequent app open $\leq$ 1 s.
Acceptance Test	Manual test on a Tecno Spark 8 mid-range Android device with network throttled via Chrome DevTools or a hardware throttler; measurement repeated on 5 reboots.

NFR-003 — API Read Response Time

Priority	MUST
Category	Performance
Description	Core API read endpoints (auth, person, ward, village, leader directory) shall respond within 400 ms for 90 percent of requests under normal load.
Measurable Target	P90 $\leq$ 400 ms; P95 $\leq$ 800 ms under 200 concurrent virtual users for 5 minutes.
Acceptance Test	k6 load test on staging environment with seeded data; results

	compared against prior baseline.
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NFR-004 — API Write Response Time

Priority	MUST
Category	Performance
Description	API write endpoints (visit log, report submission, leader add) shall respond within 800 ms for 90 percent of requests under normal load.
Measurable Target	P90 ≤ 800 ms; P95 ≤ 1500 ms under 200 concurrent virtual users.
Acceptance Test	k6 load test with mixed read/write profile; results archived.

NFR-005 — Heatmap Render Time

Priority	MUST
Category	Performance
Description	Heatmap visualisations (A, B, C) shall render in under 3 seconds for the full constituency.
Measurable Target	Full-constituency heatmap render ≤ 3 s on a 1920×1080 Chrome desktop with seeded data of 5,000 visits.
Acceptance Test	Lighthouse performance audit + manual stopwatch verification on staging.

NFR-006 — Concurrent Users — Campaign Season

Priority	MUST
Category	Performance
Description	The system shall support 200 concurrent active users during normal campaign operations with no degradation of NFR-001 through NFR-005.
Measurable Target	200 concurrent users sustained for 30 minutes with zero 5xx errors and all latency NFRs met.
Acceptance Test	k6 load test at staging; PagerDuty error budget verification.

NFR-007 — Concurrent Users — Election Day Peak

Priority	MUST
Category	Performance

Description	NyaliTrack shall support 500 concurrent active users on election day, including all polling agents, ward coordinators, and the war room.
Measurable Target	500 concurrent users sustained for 4 hours with zero 5xx errors; report submission latency P95 $\leq$ 1.5 s.
Acceptance Test	Load test at staging at 80, 100, and 120 percent of expected D-Day load; dress rehearsal at 100 percent with real coordinators.

4.2 Security NFRs

NFR-010 — Transport Encryption

Priority	MUST
Category	Security
Description	All API communication shall use TLS 1.3 or later, mitigating man-in-the-middle attacks and passive surveillance.
Measurable Target	TLS 1.3 enforced; TLS 1.2 permitted only as fallback; all earlier versions disabled. SSL Labs grade A or higher.
Acceptance Test	SSL Labs scan of production endpoints; CI gate on TLS configuration drift; archived monthly.

NFR-011 — Encryption at Rest

Priority	MUST
Category	Security
Description	All data at rest shall be encrypted using AES-256, mitigating data exposure from disk-level compromise.
Measurable Target	AES-256 encryption enabled on all database storage volumes and file storage buckets; encryption keys managed by cloud provider KMS.
Acceptance Test	Quarterly audit of cloud storage configuration; results archived.

NFR-012 — Column-Level Encryption for High-Sensitivity Fields

Priority	MUST
Category	Security
Description	National ID numbers, sensitive strategic notes on leaders and opposition, and TOTP secrets shall be encrypted at the column level with keys separated from the database.
Measurable Target	pgcrypto column-level encryption verified on: people.national_id,

	community_leaders.sensitive_notes, auth_credentials.totp_secret, and the opposition_candidates.strategic_notes field.
Acceptance Test	Manual inspection of database schema; SQL spot-check returns ciphertext, not plaintext.

NFR-013 — Authentication Rate Limiting

Priority	MUST
Category	Security
Description	Authentication endpoints shall be rate-limited to 5 requests per minute per phone/email, mitigating credential stuffing and brute force.
Measurable Target	6th authentication attempt within 60 seconds returns HTTP 429; account lockout after 5 consecutive failures.
Acceptance Test	Automated test in CI: 10 attempts in 30 seconds; assert 5 succeed/fail authentication and remainder return 429.

NFR-014 — API Rate Limiting

Priority	MUST
Category	Security
Description	API requests shall be rate-limited to 60 per minute per user and 1,000 per minute per IP, mitigating scraping and denial-of-service attempts.
Measurable Target	61st request from a single user in 60 seconds returns HTTP 429; 1001st request from a single IP returns HTTP 429.
Acceptance Test	Automated test in CI; production monitored via Redis rate-limit counters.

NFR-015 — Third-Party Security Audit

Priority	MUST
Category	Security
Description	The system shall undergo a third-party security audit before any voter-level data is loaded into production.
Measurable Target	Audit completed by a CREST-accredited or equivalent firm; report shows zero open Critical or High findings.
Acceptance Test	Audit report archived; remediation tracked in issue tracker; sign-off by Technical Lead and Founding Partner.

NFR-016 — No AI/LLM Data Transmission

Priority	MUST
Category	Security
Description	The system shall not transmit personal data, voter records, or strategic intelligence to any third-party AI or large-language-model service, mitigating data exposure via foreign-jurisdiction processing.
Measurable Target	Code search for OpenAI, Anthropic, Google Generative AI, and similar SDKs returns no references in the production codebase.
Acceptance Test	CI grep gate on the production branch; results archived per build.

4.3 Reliability & Availability NFRs**NFR-020 — Campaign-Season Uptime**

Priority	MUST
Category	Reliability
Description	The platform shall maintain at least 99.5 percent monthly uptime during the campaign season (Months 1–11).
Measurable Target	Monthly uptime $\geq$ 99.5 percent (maximum 3 hours 39 minutes downtime per month).
Acceptance Test	UptimeRobot or equivalent monitoring of public API and Central Command; monthly report archived.

NFR-021 — Election-Day Uptime

Priority	MUST
Category	Reliability
Description	NyaliTrack shall maintain at least 99.95 percent uptime during the 72-hour window surrounding election day.
Measurable Target	Uptime $\geq$ 99.95 percent over the period 06:00 8 August 2027 to 06:00 11 August 2027 (maximum 2 minutes 9 seconds downtime).
Acceptance Test	Synthetic monitoring at 30-second intervals on the NyaliTrack endpoints; report produced 11 August 2027.

NFR-022 — Database Backups

Priority	MUST
Category	Reliability

Description	The database shall be backed up automatically every 6 hours with a 30-day retention and offsite replication.
Measurable Target	Verified backup created every 6 hours; 30 days retained; replication to a second AWS region or African provider.
Acceptance Test	Quarterly restore drill from a backup; restored database opens and serves API requests successfully.

NFR-023 — Point-in-Time Recovery

Priority	MUST
Category	Reliability
Description	The database shall support point-in-time recovery to any moment in the last 7 days.
Measurable Target	PITR verified by restoring to a specified timestamp within the last week; restored data matches expected state.
Acceptance Test	Documented PITR drill performed before Months 6 and 11.

NFR-024 — Disaster Recovery Test

Priority	MUST
Category	Reliability
Description	Disaster recovery shall be successfully demonstrated at least twice before election day.
Measurable Target	Two full DR drills completed in Months 9 and 11, each restoring to a working state in under 2 hours.
Acceptance Test	DR drill reports archived; sign-off by Technical Lead.

4.4 Usability NFRs

NFR-030 — Field App Visit Logging Tap Count

Priority	MUST
Category	Usability
Description	Logging a visit in the Field App shall take three taps or fewer from app open.
Measurable Target	95 percent of test users complete visit logging in ≤ 3 taps in their first session.
Acceptance Test	Usability test with 5 ward coordinators on real Android devices; tap counts recorded and archived.

NFR-031 — Bilingual Interface

Priority	MUST
Category	Usability
Description	All interfaces (Central Command, Field App, NyaliTrack) shall support both English and Swahili, with toggle in user settings.
Measurable Target	Every user-facing string available in both languages; toggle persists across sessions.
Acceptance Test	Language coverage audit before release; review of all strings by a Swahili-native speaker.

NFR-032 — Form Validation Clarity

Priority	MUST
Category	Usability
Description	Form validation errors shall name the failed field and suggest corrective action.
Measurable Target	100 percent of validation messages reference the specific field and corrective action; no message simply says "invalid input".
Acceptance Test	UX review of all forms; recorded in /docs/ux-audit.md.

4.5 Accessibility NFRs

NFR-040 — WCAG 2.1 Level AA Conformance

Priority	SHOULD
Category	Accessibility
Description	Central Command shall meet WCAG 2.1 Level AA conformance where reasonably achievable, prioritising keyboard navigation, screen reader support, and colour contrast.
Measurable Target	Lighthouse accessibility score ≥ 90 ; manual screen reader test passes for all primary navigation flows.
Acceptance Test	Lighthouse run in CI; manual NVDA or VoiceOver test recorded before each major release.

4.6 Operational NFRs

NFR-050 — Audit Log Append-Only

Priority	MUST
Category	Operational

Description	The audit log shall be append-only; no row may be deleted or updated after insertion, mitigating insider data tampering.
Measurable Target	Database-level constraint or trigger prevents UPDATE and DELETE on the audit_log table for all roles except a sealed-key administrator.
Acceptance Test	Penetration test attempt to modify audit rows fails with permission denied; verified quarterly.

NFR-051 — Centralised Logging

Priority	MUST
Category	Operational
Description	All application, authentication, and audit logs shall be centralised in a single log management system with 7-year retention for audit logs and 90 days for application logs.
Measurable Target	CloudWatch, Loki, or equivalent receives logs from all services; retention policies verified.
Acceptance Test	Configuration audit; sample log queries return expected data.

NFR-052 — PII Redaction in Application Logs

Priority	MUST
Category	Operational
Description	Application logs shall not contain phone numbers, national IDs, voter IDs, passwords, TOTP secrets, or other personally identifying data beyond user UUIDs.
Measurable Target	Automated scan of log samples returns no matches for phone number or national ID patterns.
Acceptance Test	Daily scan in CI; alert on any match.

NFR-053 — Monitoring and Alerting

Priority	MUST
Category	Operational
Description	The system shall be continuously monitored with alerting for API 5xx rate, latency degradation, queue depth, and authentication anomalies.
Measurable Target	Alerts dispatched within 5 minutes of threshold breach to on-call

	rotation via WhatsApp and email.
Acceptance Test	Monthly chaos drill: induced failure verified to trigger alert within window.

NFR-054 — Containerised Deployment

Priority	MUST
Category	Operational
Description	All services shall be containerised (Docker) with reproducible builds.
Measurable Target	Every service has a Dockerfile committed to the repository; CI builds images on every merge to main.
Acceptance Test	Container build verification in CI pipeline.

5. External Interface Requirements

5.1 Third-Party Integrations

GUIDANCE <i>Every external service is listed with its failure-handling strategy. The fourth column is mandatory.</i>			
System	Integration Type	Purpose	Failure Handling
WhatsApp Business API (via 360dialog or Africa's Talking)	REST API + Webhook	Meeting notifications, urgent broadcasts, field comms	Exponential backoff 3 retries; SMS fallback via Africa's Talking; alert to comms_head on second failure
Africa's Talking SMS	REST API	SMS fallback for notifications; inbound short code reports on D-Day	Status polling; alert on >5 percent failed delivery; queued retry up to 1 hour
Resend or Postmark	REST API	Transactional email — meeting invitations, system notifications, security alerts	Retry on 5xx; surface failure to admin within 5 minutes; suppressions tracked
MapLibre + OpenStreetMap tiles	Tile fetch	Map rendering for Central Command, Field App, NyaliTrack	Local cache; tile load failure degrades to grey base layer with overlay still visible
Mapbox tiles (optional)	Tile fetch	Higher-quality map rendering if budget permits	Fallback to OpenStreetMap if Mapbox unreachable
Supabase Storage or AWS S3	SDK + REST	Photos for activities, team, leaders, sites	Resume-capable upload; failed uploads queued in Field App until success
Expo Push Notification Service	REST API	Mobile push notifications	Status logging; user receives in-app notification on next app open as fallback
Cloud KMS (AWS or African provider)	SDK	Key management for column-level encryption	Service degradation prevents new writes to encrypted columns; reads continue from cached key

5.2 Technology Stack

GUIDANCE <i>Every choice is justified by a one-line rationale. The rationale matters more than the choice; future readers need to understand why.</i>		
Layer	Technology	Rationale
Central Command frontend + API	Next.js 14 (App Router) + TypeScript	Team's strongest stack; SSR for fast first paint, route handlers for the API, single codebase
UI components	Tailwind CSS + shadcn/ui	Rapid composition; consistent design tokens; minimal bundle weight
Field App and NyaliTrack	Expo SDK 51+ with React Native	Native performance; reliable offline; push notifications; OTA JS updates; same React paradigm as web
Database	PostgreSQL 15 + PostGIS 3.3 + Drizzle ORM	Spatial queries are core to the platform; PostGIS is mature; Drizzle provides type-safe SQL
Local mobile DB	Expo SQLite + Drizzle	Production-grade local persistence; same ORM as the server reduces context switching
Authentication	Auth.js (web) + JWT in Expo Secure Store (mobile)	Auth.js is the de-facto Next.js auth; Secure Store provides hardware-backed token storage on mobile
Real-time sync	Supabase Realtime or Pusher	Avoids implementing WebSocket infrastructure from scratch
Caching and rate limiting	Redis 7	Standard tool for session, cache, and rate-limit counters
Map rendering	MapLibre GL JS (web) + react-native-maps (mobile)	Open-source; no vendor lock-in; OpenStreetMap tiles free at low volume
Charts	Visx or Observable Plot	Handles dense electoral data better than Chart.js
Hosting (API + DB)	AWS Cape Town (af-south-1) or Nairobi-based provider	African data residency per Constraint CON-001 and DPA §50
Mobile build and deploy	EAS Build + EAS Update	Reproducible builds; OTA JS updates without store review
Email	Resend	Modern API; reliable Kenyan delivery
SMS	Africa's Talking	Best Kenya delivery rates; mature short code support

WhatsApp BSP	360dialog or Africa's Talking	Both have Kenya operations and political-account experience
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5.3 Communication Standards

- All API communication shall use HTTPS with TLS 1.3 or later (NFR-010).
- All API endpoints shall return JSON with the response envelope: { success: boolean, data?: any, error?: { code: string, message: string } }.
- All datetimes are ISO 8601 in UTC; client converts to local time for display.
- All identifiers are UUIDv7 to support sortable, distributed generation across offline clients.
- Pagination uses cursor-based opaque tokens, not offset-limit, to remain stable as data grows.
- Webhook payloads are signed with HMAC-SHA256; consumers verify before processing.

6. System Requirements

6.1 Development Environment

Component	Specification
Version Control	Git on GitHub (private repository); feature-branch workflow
Branching Convention	main (protected), feat/*, fix/*, docs/*, hotfix/*
Commit Standard	Conventional Commits (feat:, fix:, docs:, chore:, refactor:, test:)
CI/CD	GitHub Actions; build, lint, test, type-check on every push; deploy to staging on merge to main
Code Quality	ESLint (Airbnb base + custom), Prettier, Husky pre-commit hooks
Testing Framework	Vitest for unit tests; Playwright for end-to-end; Detox for Expo E2E; k6 for performance
Local Development	Docker Compose for Postgres, Redis, MailHog; .env.example committed; .env.local in .gitignore
API Documentation	OpenAPI 3.1 spec generated from code; hosted on a private docs site
PR Review	Minimum 2 approvals before merge; 1 approval for hotfix branches; tech lead must approve schema migrations

6.2 Production Environment

Component	Specification
Hosting Platform	AWS Cape Town (af-south-1) — API and database; Vercel — Central Command web
Frontend Deployment	Vercel (Central Command); EAS Build + Play Store + App Store (Field App and NyaliTrack)
Server Minimum Spec	4 vCPU, 8 GB RAM, 100 GB SSD (auto-scaled during D-Day window)
Database	PostgreSQL 15 (managed RDS or equivalent), 100 GB initial storage, point-in-time recovery enabled
Cache	Redis 7 (managed ElastiCache or equivalent), 2 GB memory
Secrets Management	AWS Secrets Manager (or provider equivalent); never in Git
Domain / SSL	alfayonelson.campaign domain (or chosen variant); Let's Encrypt or AWS Certificate Manager

Component	Specification
Monitoring	CloudWatch + UptimeRobot for synthetic monitoring; Sentry for error tracking
Logging	JSON structured logs; CloudWatch Logs; 7-year retention for audit logs, 90 days for application logs (NFR-051)
Backups	RDS automated 6-hourly snapshots, 30-day retention, cross-region replication (NFR-022)
Disaster Recovery	RTO 2 hours, RPO 6 hours; documented runbook; drilled twice before election day (NFR-024)

7. MVP Scope & Requirements Traceability

GUIDANCE

Two tables. The first is the engineering build list. The second is the deferred backlog.

7.1 MVP Feature Set

The following features are MUST for v1.0 and shall be operationally usable before the campaign's public launch milestone (target: Month 4).

Feature	Description	Requirement IDs	Priority
Identity & Access	Authentication with 2FA; RBAC enforced by RLS	FR-001 – FR-004, NFR-010 – NFR-016	MUST
Geographic Intelligence	Constituency, ward, village, polling-station data and clickable map	FR-010 – FR-012	MUST
Community Intelligence	Leader and site directories with review queue	FR-020 – FR-022	MUST
Issue Management	Village-level issue capture with controlled vocabulary	FR-030	MUST
Activity Management	Lifecycle for rallies, barazas, visits, courtesy calls	FR-040	MUST
Visit Logging & Heatmaps	Three-tap mobile logging; Coverage, Influence, Persuasion heatmaps	FR-050 – FR-053	MUST
Meeting Coordination	Creation with WhatsApp + email notifications and reminders	FR-060	MUST
Pollings & Analysis	Electoral history, demographics, race intelligence	FR-070 – FR-072	MUST
Team Management	Org chart and directory with activity status	FR-080	MUST
Homepage & Countdown	Constituency map homepage and persistent countdown	FR-090 – FR-092	MUST

Feature	Description	Requirement IDs	Priority
Field App Foundations	Offline-first, role-aware home, push notifications	FR-100 – FR-102	MUST
Search & Export	Global search; PDF and CSV export	FR-110 – FR-111	MUST/ SHOULD
NyaliTrack Outline	Polling station board, incident reports, SMS fallback	FR-120 – FR-122	MUST (delivered Months 8–11)
Committed Supporter Network	Restricted-access registry of committed supporters with consent, audit, and withdrawal	FR-130 – FR-134, COMP-010	MUST
Performance Targets	Load, latency, capacity thresholds	NFR-001 – NFR-007	MUST
Reliability	Uptime, backups, DR drills	NFR-020 – NFR-024	MUST
Usability	Three-tap mobile flows; bilingual UI	NFR-030 – NFR-032	MUST
Operational	Append-only audit, centralised logs, PII redaction, monitoring, containers	NFR-050 – NFR-054	MUST

7.2 Post-MVP Roadmap (v1.1+)

Feature	Notes
Biometric mobile unlock	FR-004 (SHOULD) — included if Month 1 timeline permits; otherwise v1.1
Accessibility WCAG 2.1 AA full conformance	NFR-040 — best-effort in MVP; full conformance audit in v1.1
Multi-constituency deployment	Out of scope v1.0; database supports it; targeted for 2032 cycle reuse
Voter persuasion scoring (computed per voter)	Requires extended legal review; deferred to v1.2 if at all
Native canvassing route optimisation	Useful but complex; deferred to v1.1 after observing real canvasser patterns
Public candidate website	Out of scope; separate marketing site

Feature	Notes
integration	
Donor management and finance tracking	Explicitly out of scope; separate system per Section 1.2
Sentiment analysis on social media	Explicitly out of scope per Section 1.2
Direct IEBC results API integration	No such API exists; deferred indefinitely
SACCO and chama loan history overlays	Politically sensitive; deferred to v1.2 pending legal review

8. Compliance & Legal Requirements

8.1 Regulatory Compliance

- Kenya Data Protection Act, 2019 — specifically §18 (registration of data controllers), §25 (lawful basis), §26 (rights of data subjects), §40 (data protection impact assessment), §50 (data localisation), and §63 (penalties).
- Data Protection (General) Regulations, 2021 — operational rules implementing the DPA.
- Constitution of Kenya, 2010 — Article 31 (right to privacy), Articles 81–88 (electoral framework).
- Elections Act, 2011 (revised editions) — campaign conduct rules.
- Election Offences Act, 2016 — applicable to any election-day reporting features.
- Computer Misuse and Cybercrimes Act, 2018 — applicable to platform integrity and unauthorised access.

8.2 Data Privacy Requirements

- COMP-001 — Data Minimisation: only data necessary for the stated campaign purposes is collected; the platform does not collect biometric data, financial data, or sensitive medical data.
- COMP-002 — Data Localisation: all personal data is stored on infrastructure physically located in Africa, preferring Kenyan providers where feasible (Constraint CON-001).
- COMP-003 — Lawful Basis: documented lawful basis per category of personal data; voter register data processed on the basis applicable under DPA §25 as advised by engaged legal counsel.
- COMP-004 — User Consent: meeting invitees and notification recipients have explicit consent recorded; community-leader contact-record creation requires the owning team member to confirm consent.
- COMP-005 — Data Subject Rights: the platform supports data subject access requests (DSAR) per DPA §26; documented procedure for export within 30 days; rectification and erasure procedures documented.
- COMP-006 — Third-Party Data Sharing: every vendor processing personal data has a signed Data Processing Agreement; vendors are: WhatsApp BSP, email provider, SMS gateway, cloud host, file storage. No data is shared with political parties or third parties outside this list without explicit campaign-leadership authorisation.
- COMP-007 — Data Retention: campaign operational data retained until 12 months after election; audit logs retained 7 years (DPA audit obligations); WhatsApp messaging metadata retained 90 days.
- COMP-008 — Data Protection Impact Assessment: DPIA produced by Month 4 covering voter register processing, community-leader profiling, and the persuasion-opportunity computation; revisited if data scope changes.
- COMP-009 — Complaints Procedure: documented user-facing procedure for lodging complaints with the campaign's Data Protection Officer and onward to the ODPC.
- COMP-010 — Committed Supporter Data Safeguards: the Committed Supporter Network (FR-130 to FR-134) processes sensitive personal data linking voters to participation in community programs, which under the DPA §44–45 requires explicit consent, elevated lawful basis, and additional safeguards. The campaign's Data Protection Officer shall confirm the lawful basis in writing before any committed supporter record is created in production. Records are restricted to a named list of roles, every access is audit-logged, and no bulk export is permitted without campaign_manager approval.

8.3 Specific Risks and Mitigations

Risk	Severity	Mitigation
Unregistered processing of voter data	Critical	ODPC registration filed before Month 5 voter-data ingestion (DEP-002)
Leak of voter contact data	Critical	Column-level encryption, RLS, audit logging, third-party security audit (NFR-015)
Leak of Committed Supporter list	Critical	Restricted-role access (FR-130 AC-130.4); audit on every read; no default export; monthly access audit (AUD-130.3); consent withdrawal procedure (FR-134)
Improper use of ethnic composition data	High	Aggregation only at ward level or higher (FR-071); no individual-level ethnicity collection
WhatsApp account suspension or ban	Medium	BSP relationship; SMS fallback (NFR-022 path)
Opposition lawsuit or injunction	Medium	Legal review of data sources before each release; engaged counsel
Insider data exfiltration	High	RLS enforcement; audit logging; minimum-necessary scoping; offboarding procedure
Election-day system overload	Critical	Load testing Months 9–10; dress rehearsals; SMS fallback (FR-122)
Network outages on D-Day	High	Offline-first Field App (FR-100); SMS short code (FR-122); paper backup forms documented
App Store rejection of political app	Medium	Early submission with appropriate metadata; Play Store generally more permissive than Apple

Document Control

Revision History

GUIDANCE		Update with every material change. Reference the change request that authorised each entry.	
Version	Date	Author	Summary of Changes
v0.1	26 May 2026	Developers Mania Technical Lead	Initial draft prepared from product owner brief and conversational requirements gathering. Covers all 8 sections, 30+ FRs, 30+ NFRs.
v0.2	26 May 2026	Developers Mania Technical Lead	Added Committed Supporter Network domain (FR-130 to FR-134) covering voters with documented support commitments arising from community programs. Added COMP-010 (lawful basis), added committed supporter leak risk to risk matrix, added domain to MVP traceability. Updated Section 1.2 scope.
v0.3	[Pending]	[Author]	[Incorporate peer review feedback]
v1.0	[Pending]	[Author]	[First signed version after Technical Lead, Account Lead, Client, and Founding Partner sign-off]

Approval Sign-Off

GUIDANCE		This is a T3+ project. Required signatures: Technical Lead, Account Lead, Client Representative, and Founding Partner.	
Role	Name	Signature	Date
Technical Lead	[To be assigned]		
Account Lead	[To be assigned]		
Client Representative (Campaign Manager)	[To be assigned]		
Aspirant	Alfayo Nelson		
Founding Partner	[To be assigned]		

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